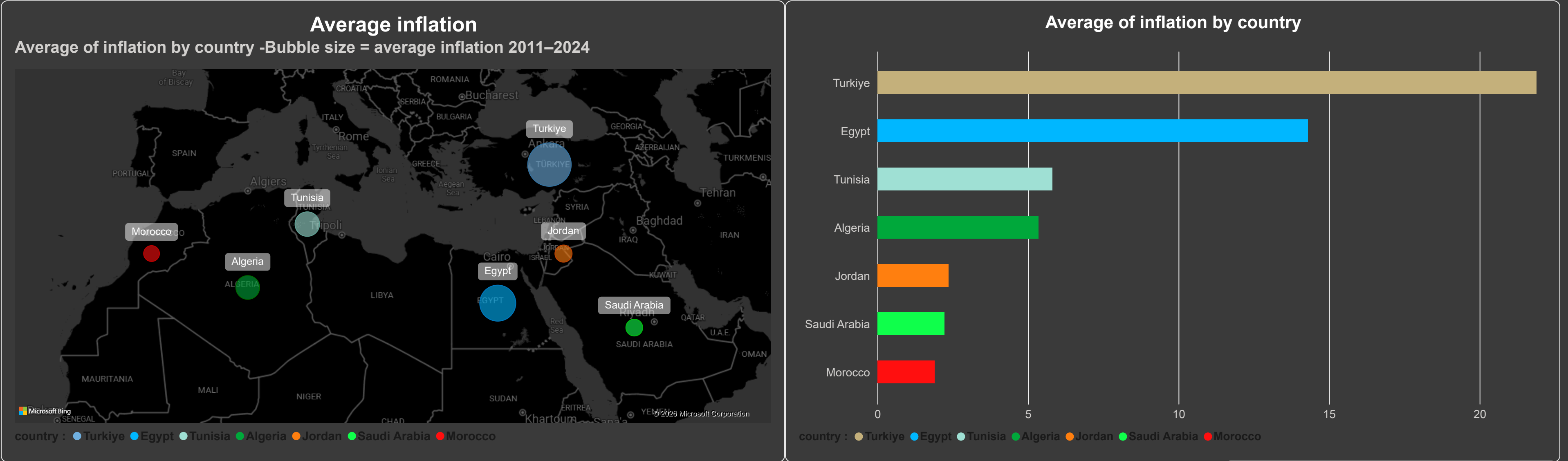




Average of inflation

7.69

Maximum annual inflation

72.31

Turkiye

Minimum annual inflation

-1.19

Morocco

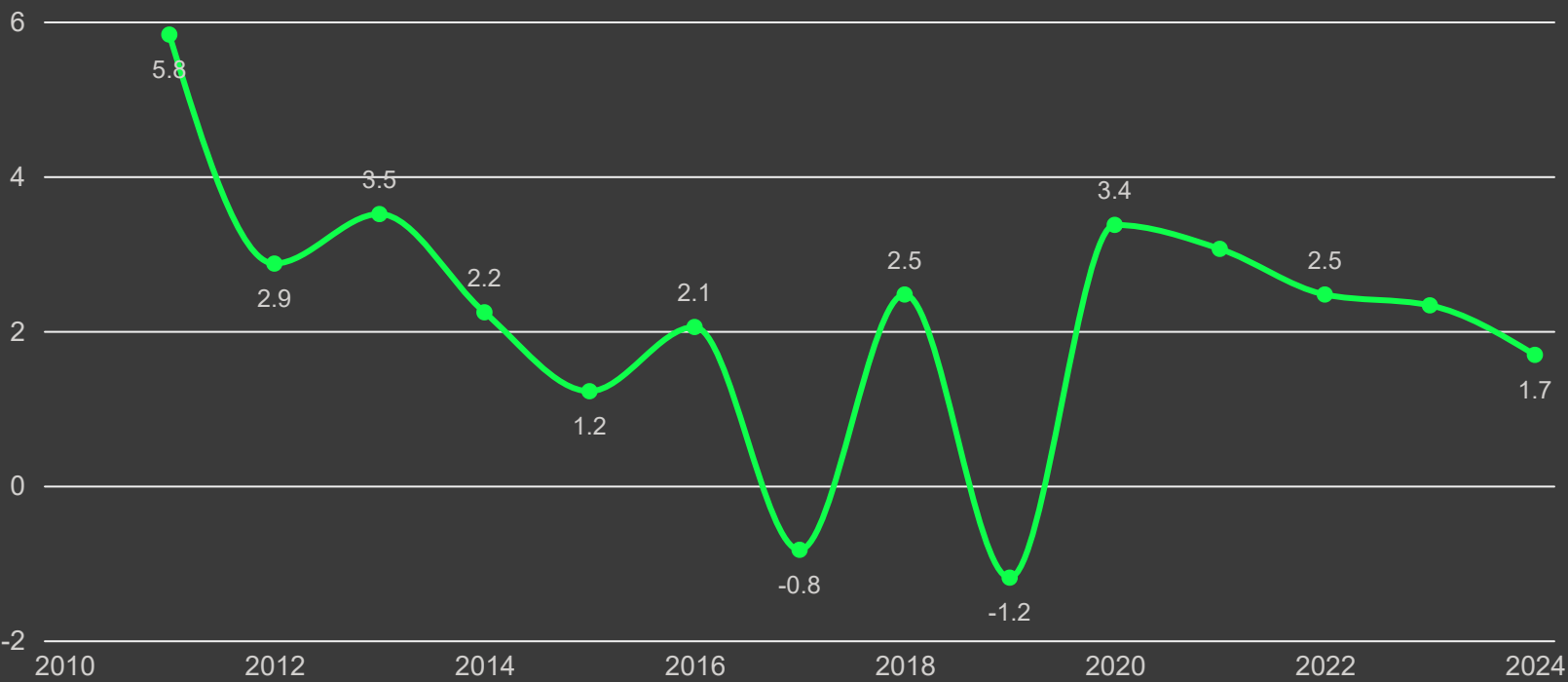
Average of Broad Money Growth

11.88

Country	Mean inflation (%)	Mean GDP Growth (%)	Mean Broad Money Growth (%)	Mean Depreciation Rate (%)
Algeria	5.35	2.37	8.82	4.50
Egypt	14.30	3.75	19.64	18.45
Jordan	2.35	2.31	3.87	0.00
Morocco	1.89	2.97	5.53	1.37
Saudi Arabia	2.22	3.87	7.39	0.00
Tunisia	5.81	1.18	8.77	5.87
Turkiye	21.88	5.57	29.13	26.00

Variable Trends Across MENA Countries (2011–2024)

Inflation (%) over time

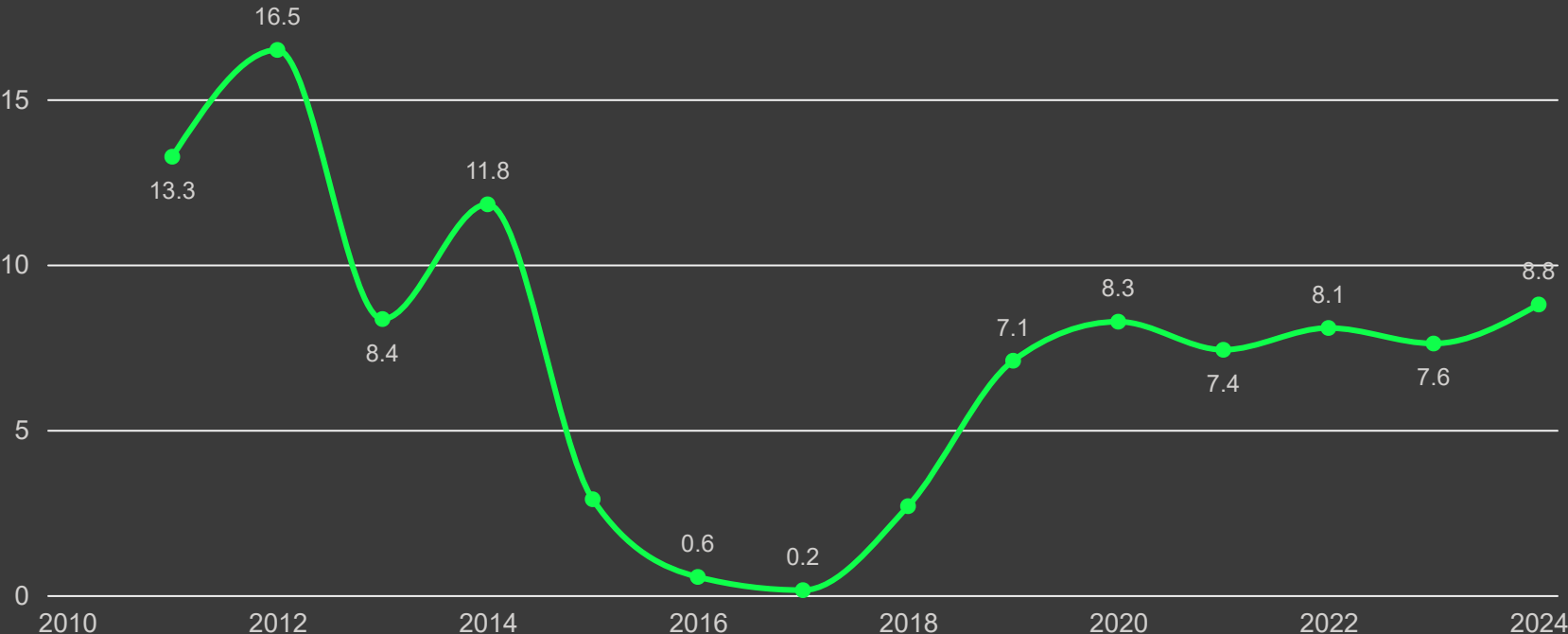


Country : Saudi Arabia

country

- ☐ Algeria
- ☐ Egypt
- ☐ Jordan
- ☐ Morocco
- ☒ Saudi Arabia
- ☐ Tunisia
- ☐ Turkiye

Broad Money Growth (%) over time

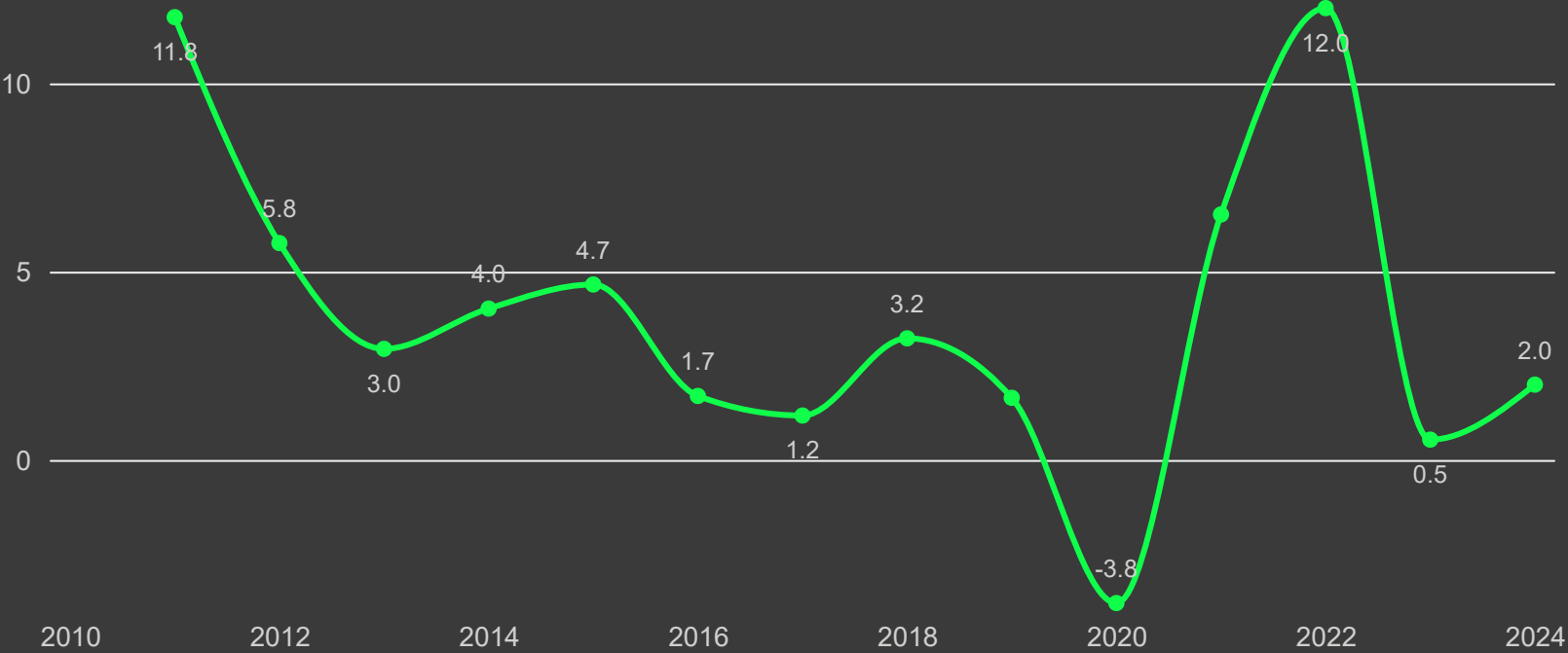


Country : Saudi Arabia

Note

Note : Saudi\Jordan = pegged 0%

GDP Growth (%) over time

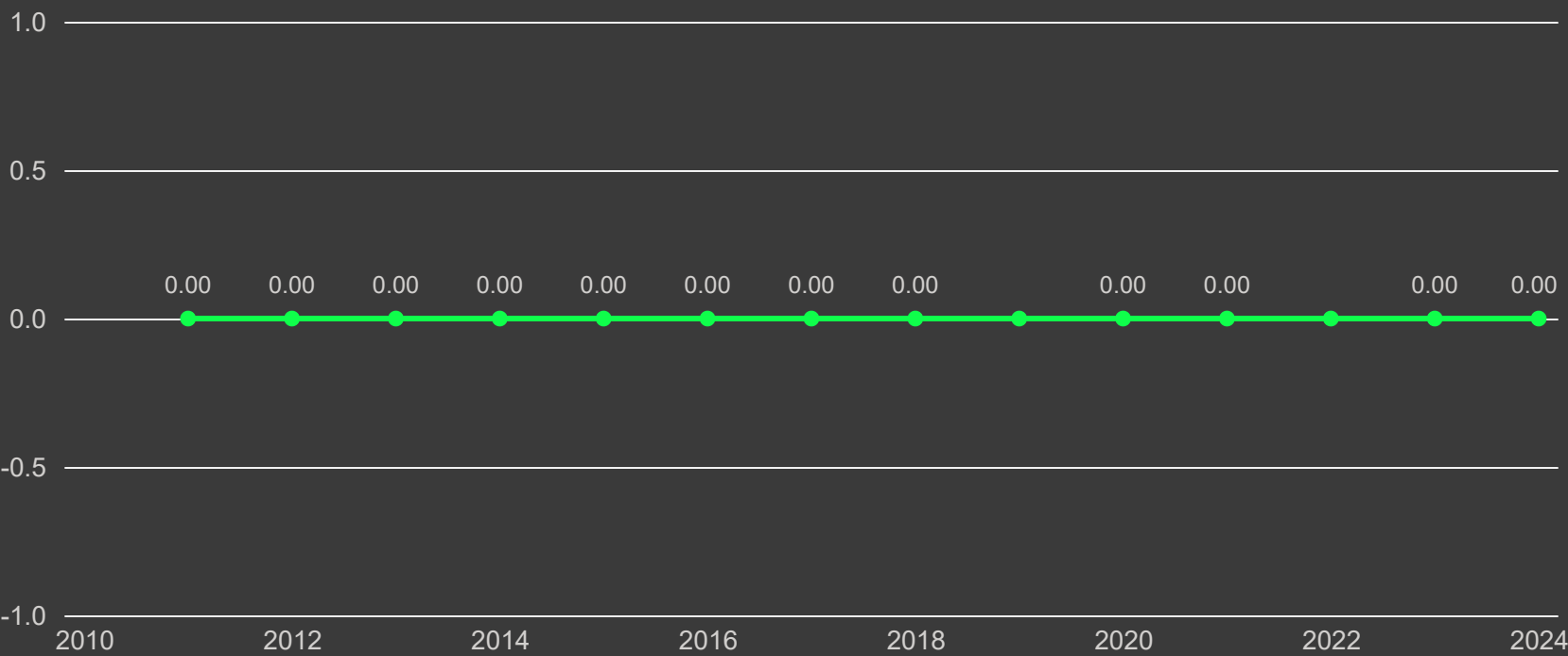


Country : Saudi Arabia

year

- ☐ 2011
- ☐ 2012
- ☐ 2013
- ☐ 2014
- ☐ 2015
- ☐ 2016
- ☐ 2017
- ☐ 2018
- ☐ 2019
- ☐ 2020
- ☐ 2021
- ☐ 2022
- ☐ 2023
- ☐ 2024

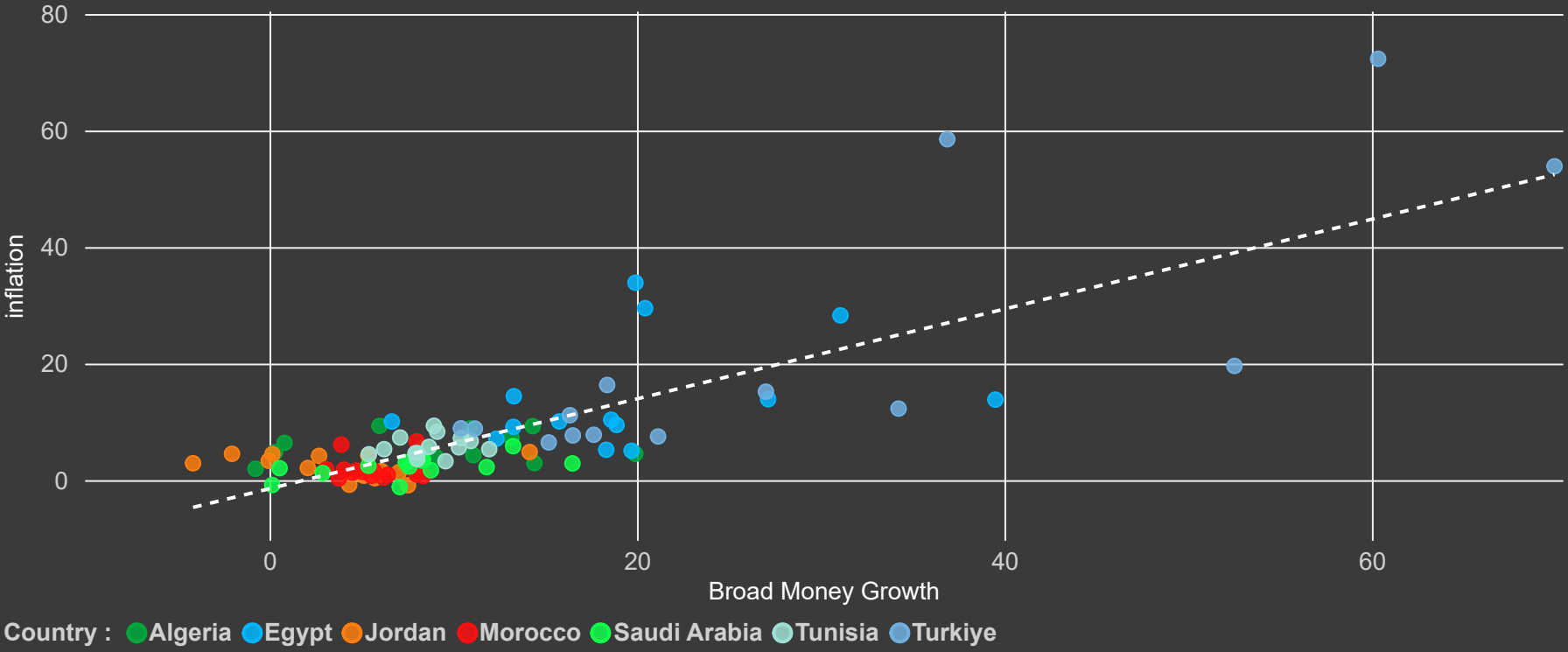
Depreciation Rate (%) over time



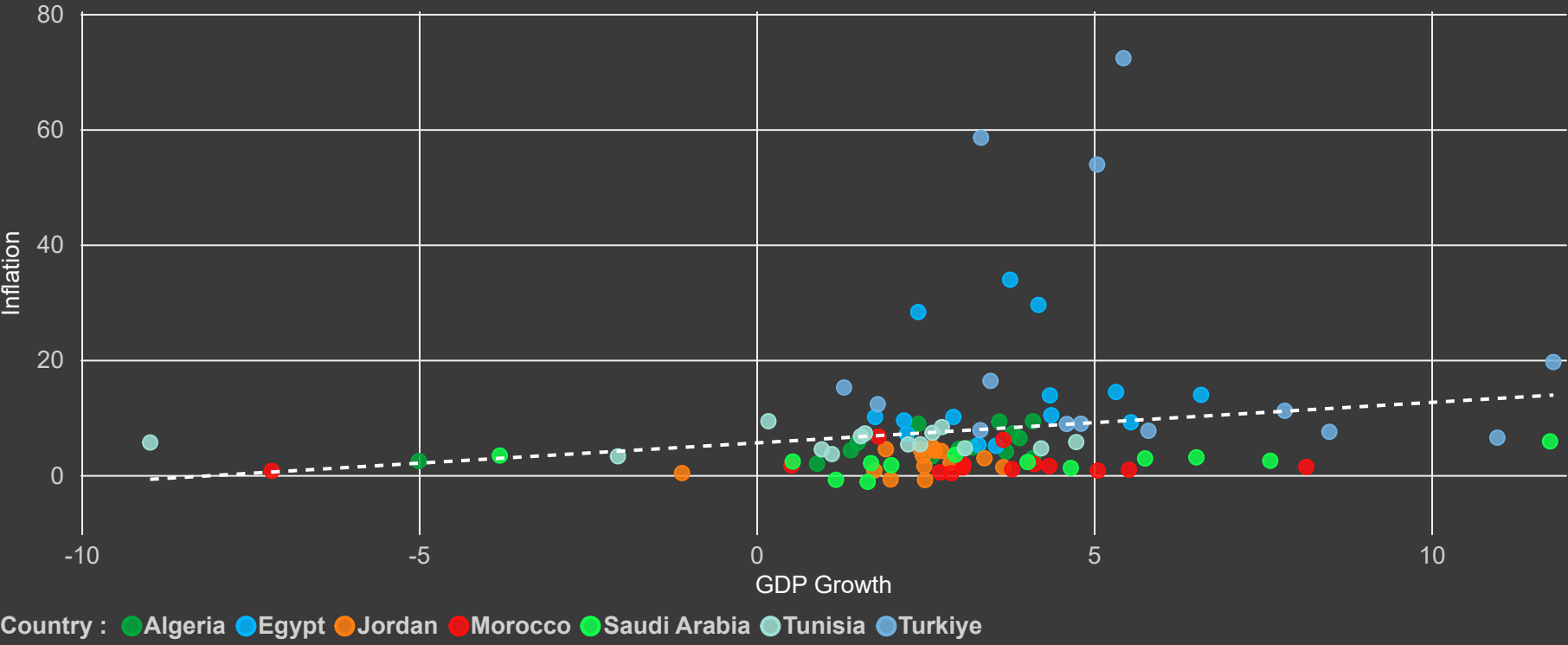
Country : Saudi Arabia

Variable Relationships — Testing the QTM Hypothesis

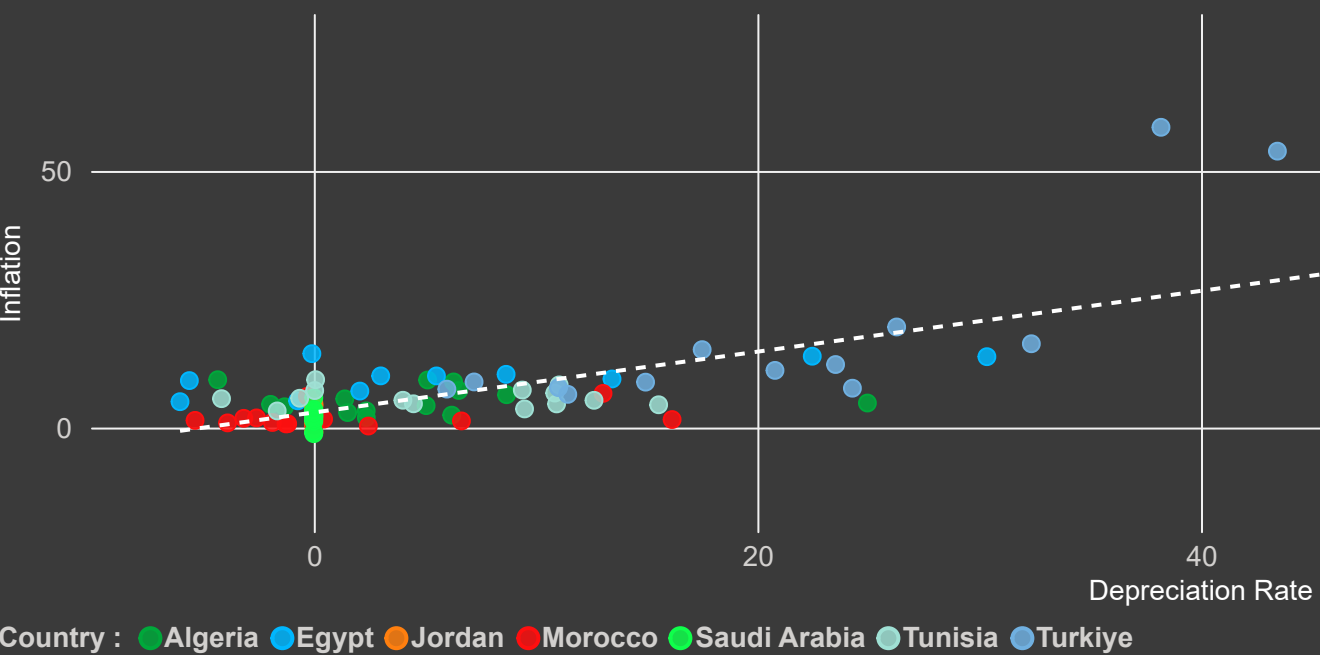
Broad Money vs Inflation



GDP Growth vs Inflation



Depreciation Rate vs Inflation



country

- Algeria
- Egypt
- Jordan
- Morocco
- Saudi Arabia
- Tunisia
- Turkiye

year

- 2011
- 2012
- 2013
- 2014
- 2015
- 2016
- 2017
- 2018
- 2019
- 2020
- 2021
- 2022
- 2023
- 2024

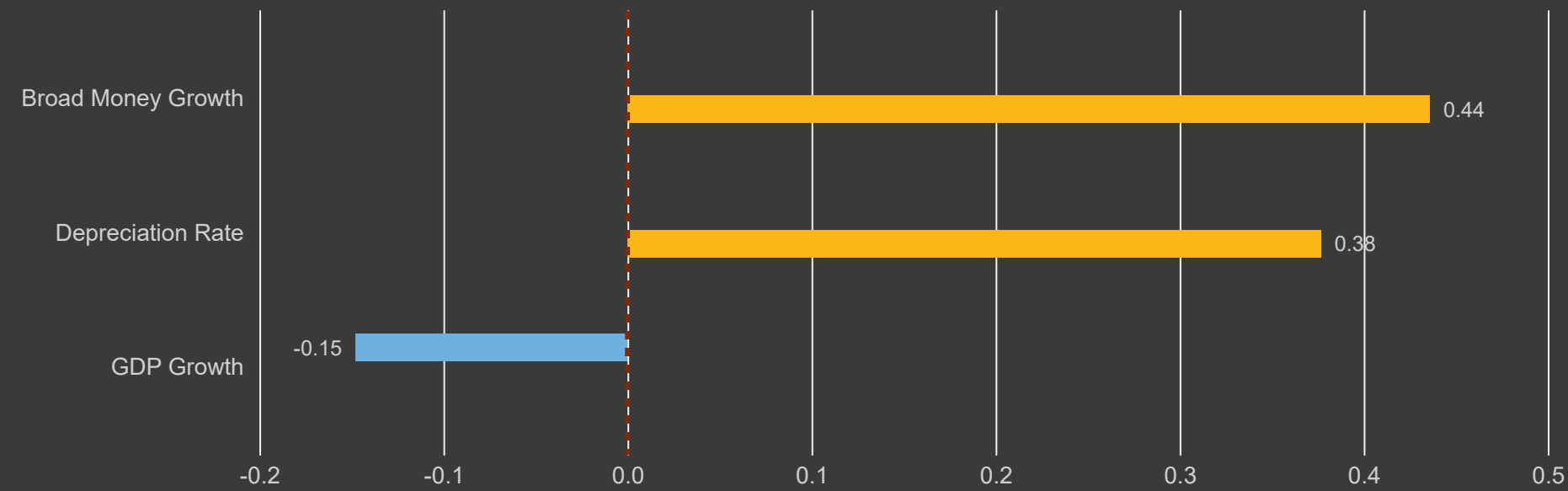
The positive relationship between broad money growth and inflation is consistent with the monetary explanation of inflation and the Quantity Theory of Money.

Results are largely driven by Turkey's extreme values. Sensitivity analysis (Page 4) confirms this.

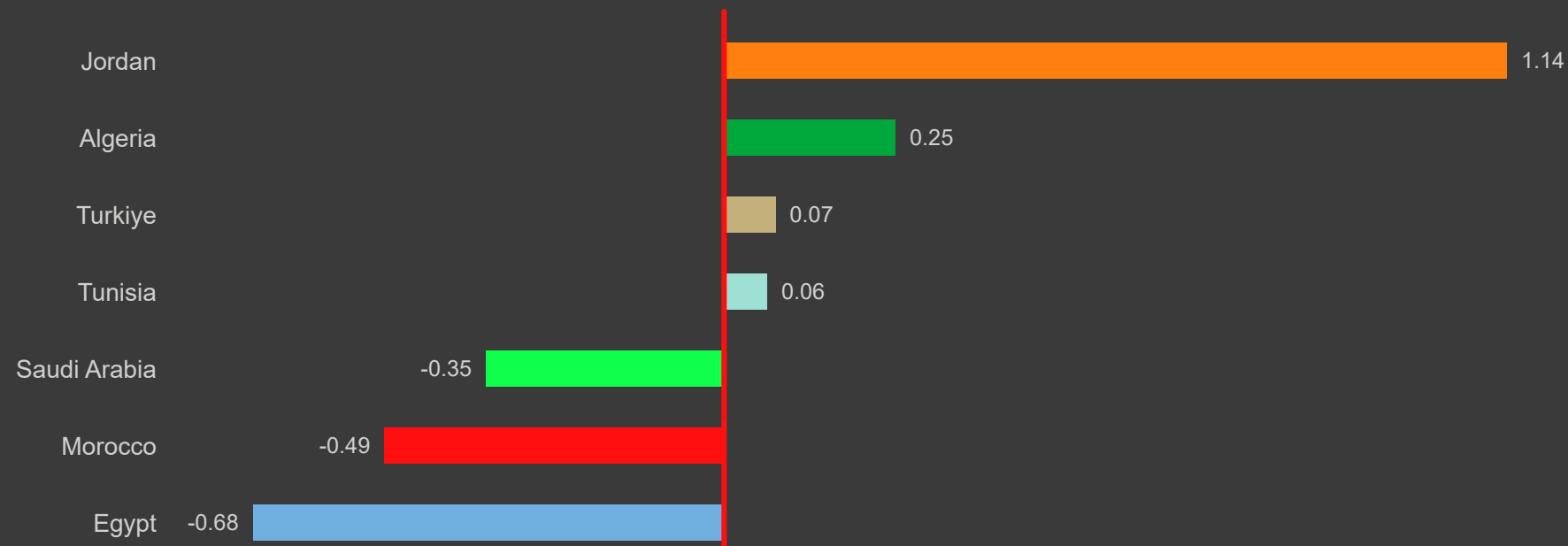


Model Results — MENA (2011–2024)

Coefficient Plot with 95% CI



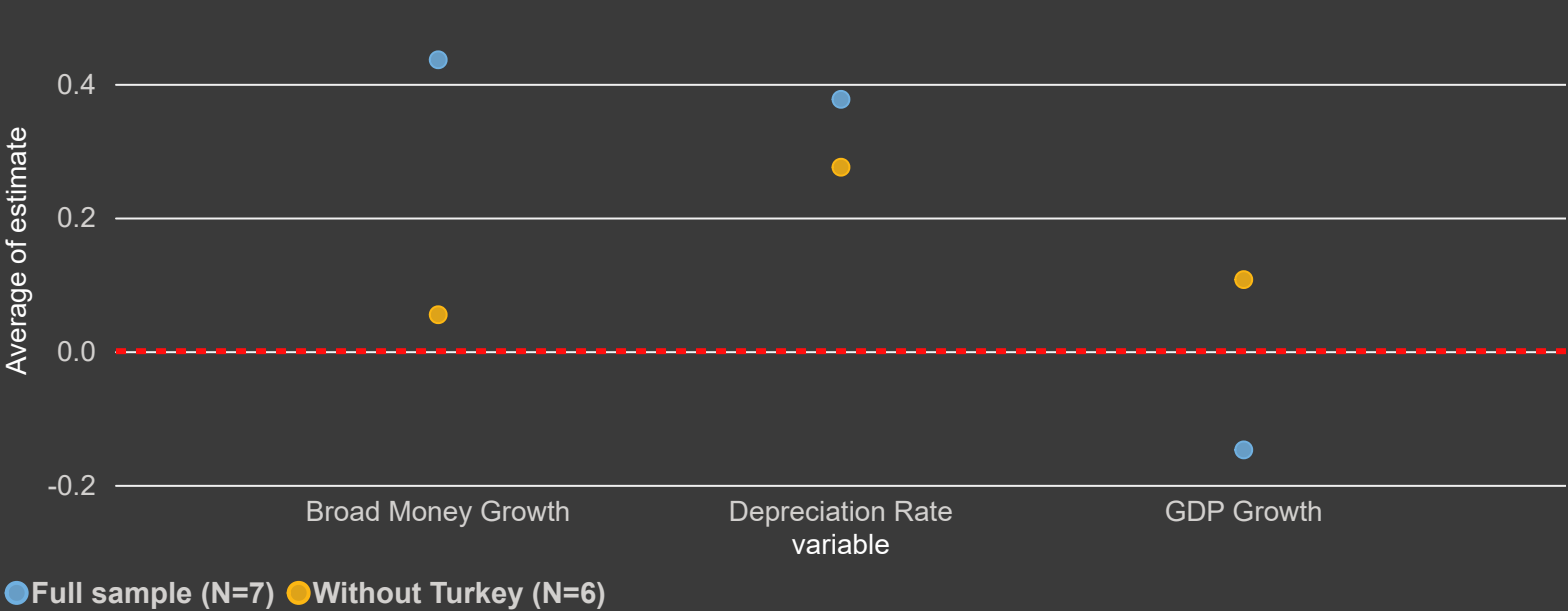
Country Fixed Effects (α_i)



Model	R2	R2 Type
FE Individual	0.68	Within
FE Two-way	0.67	Within
Pooled OLS	0.80	Overall
Random Effects	0.80	Overall

Variable	β Coefficient	Std. Error	CI Upper (95%)	CI Lower (95%)	Significance
Broad Money Growth	0.44	0.06	0.56	0.31	Significant
Depreciation Rate	0.38	0.09	0.55	0.21	Significant
GDP Growth	-0.15	0.24	0.32	-0.62	Not Significant

Sensitivity



Observations: 98

Countries: 7

Period: 2011–2024

Selected estimator: Pooled OLS

Selection:

BP-LM $p=0.10$, F-test $p=0.98$ → no panel effects

Robust SE: HC3 country-clustered + Driscoll-Kraay

Specification & Diagnostic Tests

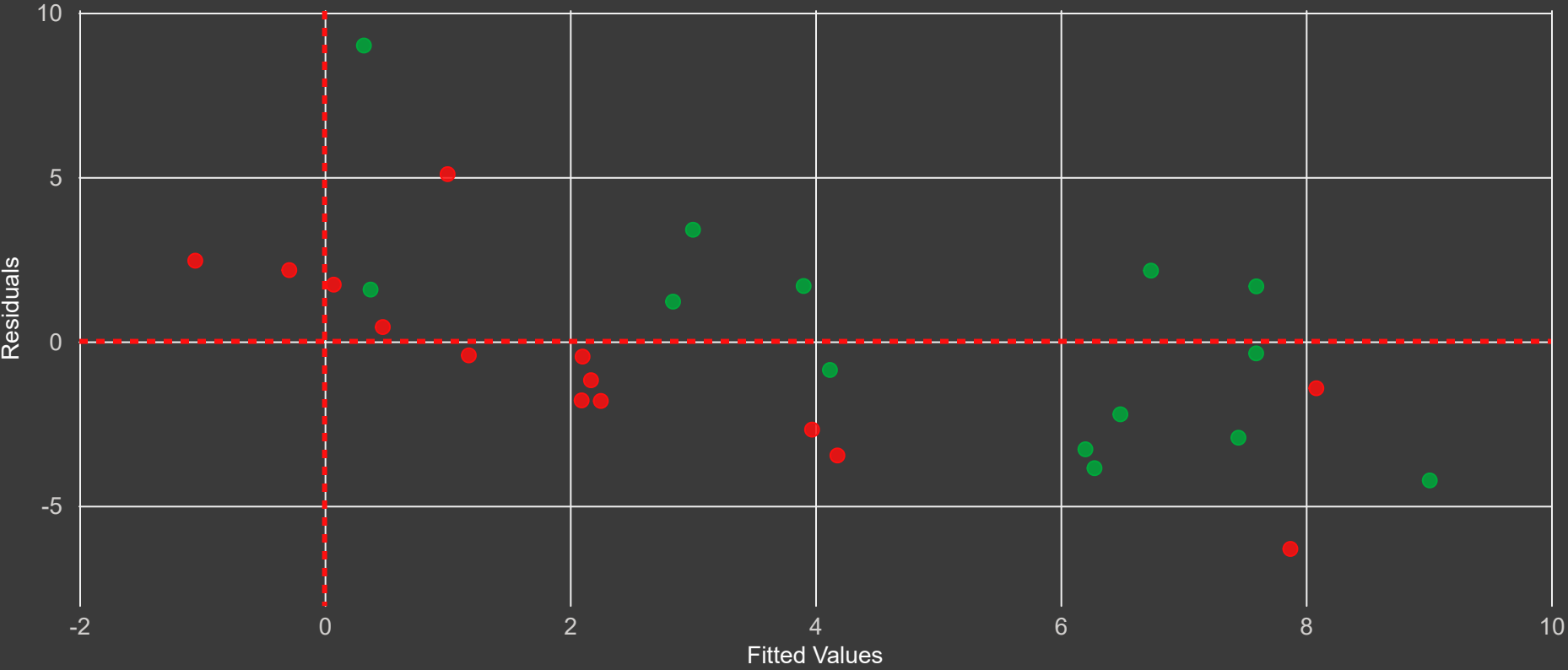
Specification & Diagnostic Test Results

Test	Null Hypothesis	P Value	Decision	Result
Autocorrelation (pbgtest)	No serial correlation	0.01	Reject H0	Significant
Breusch-Pagan LM	No individual panel effects	0.10	Do not reject H0	Not significant
Cross-Sectional Dependence	No cross-sectional dependence	0.03	Reject H0	Significant
F-test (FE vs OLS)	No individual fixed effects	0.98	Do not reject H0	Not significant
Hausman (Auxiliary)	RE consistent (robust)	1.00	Do not reject H0	Not significant
Hausman (Standard)	RE consistent	1.00	Do not reject H0	Not significant
Heteroscedasticity (BP)	Homoscedastic errors	0.00	Reject H0	Significant
Time FE Test	No time fixed effects	0.08	Do not reject H0	Not significant
White Test	Homoscedastic errors (general form)	0.00	Reject H0	Significant

country

- ☒ Algeria
- ☐ Egypt
- ☐ Jordan
- ☒ Morocco
- ☐ Saudi Arabia
- ☐ Tunisia
- ☐ Turkiye

Residuals vs Fitted values



Country : ● Algeria ● Morocco

Actuel vs Fitted Inflation

By Country

